

Laxmi Charitable Trust's
Sheth L.U.J College of Arts & Sir M.V. College of Science and Commerce
Department of Information Technology (B.Sc.I.T Semester IV)
Dot .Net Core and Progressive Web Application Practical

Practical – III

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Aim(A) : Create methods add(), multiply(), subtract() ,divide() with suitable parameters and call these methods using concept of C# delegate.

Algorithm:

Step 1 Start

Step 2: Declare the delegate : delegate void Arithmetic_cal(int a,int b);

Step 3: Create methods having the same signature

Step 4: Create a delegate object : Arithmetic_cal ac;

Step 5: Assign a method to the delegate : ac =c.add;

Step 6: Invoke the delegate : ac(10,20);

Step 7 : Stop

Code:

```
using System;
delegate void Arithmetic_cal(int a,int b);
class Calculator{
    public void add(int a, int b){
        Console.WriteLine("Additon : "+(a+b));
    }
    public void sub(int a, int b){
        Console.WriteLine("Subtraction : "+(a-b));
    }
    public void mul(int a, int b){
        Console.WriteLine("Multiplication : "+(a*b));
    }
    public void div(int a, int b){
        Console.WriteLine("Diviosn : "+(a/b));
    }
}
public class HelloWorld
{
    static void Main(string[] args)
    {
        Calculator c = new Calculator();
        Arithmetic_cal ac;
        ac =c.add;
        ac(10,20);
    }
}
```

```

        ac =c.sub;
        ac(10,20);
        ac =c.mul;
        ac(10,20);
        ac =c.div;
        ac(10,20);
    }
}

```

Output:

```

Additon : 30
Subtraction : -10
Multiplication : 200
Diviosn : 0

```

Aim(B) : Write a program using multicast delegate.

Algorithm:

Step 1 Start

Step 2: Declare the delegate : delegate void Message();

Step 3: Create methods having the same signature : public void Hello()

Step 4: Create a delegate object and assign method : Message msg = mp.Hello

Step 5: Multicast the delegate : msg +=mp.Msg;

Step 6 : Stop

Code:

```

using System;
delegate void Message();
class Msg_Print{
    public void Hello(){
        Console.WriteLine("HELLO TO C# CODING ");
    }
    public void Msg(){
        Console.WriteLine("This is your welcome Message ");
    }
    public void Thank(){
        Console.WriteLine("Thank you for Coding ");
    }
}

```

```

public class Program
{
    static void Main(string[] args)
    {
        Msg_Print mp = new Msg_Print();
        Message msg = mp.Hello;
        msg +=mp.Msg;
        msg +=mp.Thank;
    }
}

```

```
        msg();  
    }  
}
```

Output:

```
HELLO TO C# CODING :  
This is your welcome Message  
Thank you for Coding
```

Aim(C) : Create a class BankAccount with AccountNumber and Balance. Implement property for Balance with validation (Balance cannot be negative).

Algorithm:

Step 1 Start

Step 2: Declare the class :

Step 3: Create methods having the same signature and use get,set functions

Step 4: Create a delegate object :

Step 5: Assign a method to the delegate :

Step 6: Invoke the delegate :

Step 7 : Stop

Code:

```
using System;  
class Bank{  
    public int Acc_No{get;set;}  
    private double bal;  
    public double Bal{  
        get{  
            return bal;  
        }  
        set{  
            if(value >=0){  
                bal=value;  
            }  
            else{  
                Console.WriteLine("Balance cannot be Negative");  
            }  
        }  
    }  
}  
class Program  
{  
    static void Main(string[] args)  
    {  
        Bank b = new Bank();  
        b.Acc_No = 101;  
        b.Bal = 5000;  
    }  
}
```

```
Console.WriteLine("Account Number : "+b.Acc_No);  
Console.WriteLine("Balance : "+b.Bal);  
b.Bal=-1000;  
  
    }  
}
```

Output:

```
Account Number : 101  
Balance : 5000  
Balance cannot be Negative
```